
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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ABSTRACT

Mixture of Experts (MoE) architectures now underpin the scaling of contemporary large language models, yet the mechanistic principles that distinguish them from dense networks remain largely uncharacterized. Prior work has established that dense models exploit *superposition*—encoding more features than they possess dimensions—and that the extent of this superposition is jointly governed by feature sparsity and feature importance. We demonstrate that this dense-model lens does not transfer to MoEs: neither feature sparsity nor feature importance induces the discontinuous phase changes that characterize dense representations. It is instead *network sparsity*—the ratio of active to total experts—that governs the representational strategy an MoE adopts. To render this behavior measurable, we introduce metrics that quantify superposition at the granularity of individual experts, and we establish that models with greater network sparsity exhibit systematically greater *monosemanticity*. Building on this result, we advance a definition of expert specialization grounded in monosemantic feature representation rather than load balancing, and we show that experts organize around coherent feature combinations whenever initialization directs them toward those features. Taken together, these findings establish that network sparsity renders MoEs more interpretable without a commensurate loss in reconstruction performance, directly challenging the prevailing assumption that interpretability and capability stand in fundamental tension.

1 INTRODUCTION

Mixture of Experts (MoEs) have become the dominant architecture in state-of-the-art language models, including Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), principally because conditional computation delivers substantial gains in both efficiency and performance (Shazeer et al., 2017; Fedus et al., 2022). A rich body of subsequent work has refined this design space, improving routing through mechanisms such as Expert-Choice routing and stabilizing training and transfer through approaches such as ST-MoE (Zhou et al., 2022; Zoph et al., 2022). Complementary theoretical and empirical results demonstrate that learnable routers recover latent cluster structure in the data, furnishing a first account of why experts specialize (Dikkala et al., 2023). Yet despite this widespread adoption and architectural maturity, the internal representations of MoEs remain poorly understood from a mechanistic interpretability standpoint—a gap that directly motivates the present study.

A parallel line of interpretability-oriented work has sought to render expert behavior more transparent. Yang et al. (2025b) propose MoE-X, which couples sparsity-aware routing with wide, ReLU-based experts to suppress polysemanticity, while Park et al. (2025) introduce MONET, scaling the expert count so that capabilities can be edited through targeted expert activation. These contributions, however, operate principally at the level of architectural intervention: they impose interpretability from without, and consequently leave unresolved the more fundamental questions of how MoEs represent features intrinsically, how the expert structure shapes superposition, and whether experts specialize of their own accord in the absence of additional regularization. Surveying the field, Mu & Lin (2025) identify precisely

this mechanistic understanding as a central open challenge—the very question this paper resolves.

A fundamental obstacle to interpreting neural networks is the phenomenon of superposition: the regime in which a model represents more features than it has dimensions. Superposition permits a network to pack many sparse features into a smaller set of neurons, but it does so at a direct interpretive cost, rendering individual neurons polysemantic and thereby resistant to clean, feature-level analysis.

MoE architectures introduce a new axis to this problem: network sparsity. Whereas dense models activate every neuron irrespective of the input, an MoE activates only a fraction of its total parameters on any given forward pass (Shazeer et al., 2017). This distinction is representationally consequential. A dense model is compelled to exploit feature sparsity by packing many sparse features into shared neurons; an MoE, by contrast, can afford to be selective, dedicating entire experts to specific feature combinations rather than compressing them into a common basis. Network sparsity thus supplies MoEs with a representational degree of freedom that dense models simply lack, and it is this degree of freedom whose consequences we characterize.

We organize our investigation around three questions: (1) whether MoEs exhibit less superposition than their dense counterparts; (2) whether the superposition of a given feature in an MoE expert undergoes the discrete phase change—across the feature’s relative importance and the overall feature sparsity—that is observed in dense models; and (3) whether expert specialization is more faithfully understood through the lens of feature representation than through load balancing alone.

We address these questions using controlled toy models that extend the framework of Elhage et al. (2022) to the MoE setting, a design that isolates the effect of network sparsity while holding parameter budgets fixed. Our analysis yields three principal contributions: (1) we establish that, unlike dense models, MoEs do not undergo sharp phase changes, instead exhibiting increasingly continuous representational transitions as network sparsity increases; (2) we demonstrate that MoEs consistently attain greater monosemanticity—equivalently, less superposition—than dense models matched on both active and total parameters, with individual experts representing features more cleanly; and (3) we introduce an interpretability-focused definition of expert specialization grounded in monosemantic feature representation, and we show that experts organize around coherent feature combinations rather than the arbitrary load balancing assumed by conventional accounts.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis holds that networks encode concepts as directions in activation space (Park et al., 2024), yet the number of interpretable features a network must represent routinely exceeds its available dimensions. Elhage et al. (2022) formalized the resolution of this tension as superposition, establishing that dense models pack non-orthogonal features so as to maximize representational capacity at the cost of polysemanticity. Although superposition is provably efficient as a capacity strategy (Scherlis et al., 2025), it forces reliance on elaborate post-hoc disentanglement methods—most notably Sparse Autoencoders—to recover monosemantic features (Bricken et al., 2023). More recent work characterizes how data correlations shape superposition (Prieto et al., 2025) and how interference patterns arise and may be mitigated (Gurnee et al., 2023). We extend this line of inquiry to MoE architectures and establish that it is network sparsity—rather than feature sparsity alone—that governs the representational strategy a model adopts.

Interpretability of MoEs. Although MoEs have become the standard vehicle for scaling large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), mechanistic understanding of their internal representations lags well behind that of dense networks. Existing analyses concentrate on macroscopic behaviors—routing stability (Zoph et al., 2022), expert-choice statistics (Zhou et al., 2022), and latent cluster discovery (Dikkala et al., 2023)—rather than on the geometry of the representations themselves. The more

recent interpretability-focused proposals attempt to impose specialization through architectural constraints, whether predefined concept routing (Yang et al., 2025b) or expert counts scaled to match vocabulary sizes (Park et al., 2025); yet these prioritize capability editing over an account of intrinsic feature geometry. We close this gap by analyzing directly how the expert structure shapes superposition: we establish that MoEs exhibit systematically greater monosemanticity than dense models, and we advance a feature-based definition of expert specialization that connects this geometry to routing behavior.

3 BACKGROUND

A central aim of mechanistic interpretability is to reverse-engineer neural networks, and the standard approach is to decompose their representations into a set of human-interpretable concepts termed ‘features’. These features are conventionally taken to be linear, so that any hidden state h admits the decomposition

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

where $\vec{f}_i$ is the direction associated with feature i , α_i is that feature’s activation strength—roughly, the degree to which feature i is present in the input—and F denotes the set of all represented features.

The *superposition hypothesis* holds that a model can represent far more features than it has dimensions, i.e. $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). Accommodating more features than dimensions within a latent space necessarily requires that the feature vectors in F be packed non-orthogonally. When two features exhibit nonzero interference, $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$, we say they are in *superposition*. Such superposition is tolerable precisely when features are sparsely active—a defining characteristic of most online text data—but it exacts an interpretive cost: the recovered coefficient α_i then absorbs spurious activations unrelated to the actual presence of feature i in the input.

Monosemanticity is a property of individual neurons: a neuron is monosemantic when its activation corresponds cleanly to a single coefficient α_i , that is, when features are basis-aligned. Crucially, orthogonality and basis-alignment are distinct properties—when features are orthogonal (and hence not in superposition) but not basis-aligned, neurons remain *polysemantic* even though feature interference is minimal. Our focus throughout is on reducing superposition rather than on enforcing basis-alignment; accordingly, and for brevity, when we describe a model, expert, or feature as “more monosemantic” we mean precisely that it exhibits less superposition.

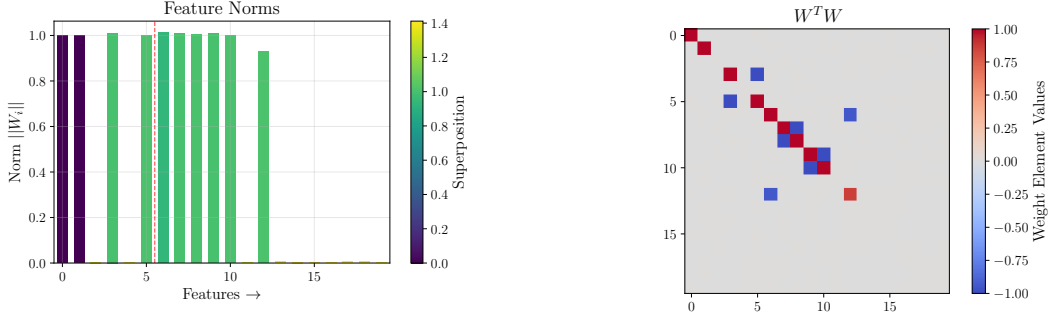
4 DEMONSTRATING SUPERPOSITION

MoEs are commonly conceptualized as compositions of dense models, with each expert treated as an independent dense network. Whether experts in fact represent features in the same manner as dense models, however, has never been established. We test this assumption directly, comparing the superposition that arises within MoEs against that of dense models under matched parameter budgets.

4.1 EXPERIMENTAL SETUP

Our objective is to characterize how an MoE projects a high-dimensional input vector $x \in \mathbb{R}^n$ into lower-dimensional expert representations $h \in \mathbb{R}^m$ and subsequently reconstructs it. This construction extends the toy-model framework of Elhage et al. (2022) to the MoE setting.

The input distribution. The input vector x models the activations of an idealized, fully disentangled model in which each dimension x_i corresponds to a distinct, independent feature—that is, a perfectly neuron-aligned, monosemantic representation. We draw $x_i \sim U(0, 1)$ subject to a prescribed sparsity $S \in [0, 1)$, so that $P(x_i = 0) = S$. Because features do not contribute equally to the reconstruction objective, we assign each feature x_i a scalar



(a) Norm of each feature’s weight vector $\|W_i\|$, with colors indicating superposition status (green for features in superposition, purple for monosemantic features).

(b) $W^\top W$ matrix where each cell represents $(\hat{W}_i \cdot W_j)$, revealing interference patterns between features.

Figure 1: Feature representation and superposition in a dense model with $n = 20$ features and $m = 6$ hidden dimensions, trained under exponentially decaying importance $I = 0.7^i$ and uniform feature density $(1 - S) = 0.1$. The superposition score encoded by color is $\sum_j (\hat{W}_i \cdot W_j)^2$, namely the total projection of all other features onto the direction of feature i ; larger values indicate that feature i shares its dimensions with more of the remaining features.

importance I_i that weights its term in the loss. To isolate and study the phase transition of a single feature, we further scale the magnitude of the last feature by a factor $r \in \mathbb{R}^+$, yielding the importance vector $I = (1, 1, \dots, r)$. The parameters I_i and r thus jointly control the signal strength of individual features, affording precise control over expert sensitivity to feature magnitude.

Model Architecture. The MoE comprises E experts, each expert e being parameterized by a weight matrix $W^e \in \mathbb{R}^{m \times n}$ and a bias $b^e \in \mathbb{R}^n$. A learned router $g(x) = \text{softmax}(W^r x)$, with $W^r \in \mathbb{R}^{E \times n}$, assigns each input to its top- k experts. Every active expert first projects the input onto a lower-dimensional hidden state $h^e = W^e x$ and then reconstructs it as $\hat{x}^e = \text{ReLU}((W^e)^\top h^e + b^e)^1$. The network output is the gate-weighted sum over the active experts, $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$, where the w_e are the renormalized gating weights.

We train each model to minimize an importance-weighted $L2$ reconstruction loss, $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$, in which the per-feature importances I_i scale the corresponding error terms. In the experiments of Section 4, and to prevent expert collapse, we augment this objective with the standard auxiliary load-balancing loss (Fedus et al., 2022), $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$, where N is the total number of experts, f_e is the fraction of batch samples routed to expert e , P_e is the mean gating probability assigned to expert e over the batch, and $\alpha = 0.01$ sets the penalty strength. We deliberately omit this auxiliary loss in Section 4 so as to isolate the MoE’s intrinsic architectural bias for or against superposition, uncontaminated by any externally imposed balancing pressure.

4.2 MEASURING FEATURE CAPACITY

To analyze feature representations across architectures, we quantify two complementary properties of each feature: its *representation strength* and its *interference* with the remaining features. Representation strength is captured by the norm $\|W_i^e\|$ of feature i ’s weight vector within expert e , which measures the extent to which that expert represents the feature: $\|W_i^e\| \approx 1$ indicates that feature i is fully represented in expert e , whereas $\|W_i^e\| \approx 0$

¹The output ReLU enforces non-negative reconstructions, consistent with our input distribution, in which $x_i \in [0, 1]$ whenever nonzero. Moreover, the off-diagonal terms of $(W^e)^\top W^e$ produce negative interference, which the ReLU suppresses; when features are sparse, this negative interference becomes effectively “free” because it is clipped to zero, which in turn incentivizes configurations with negative off-diagonal terms, such as antipodal feature pairs.

indicates that it is not learned. Interference is visualized through the Gram matrix $W^\top W$, whose off-diagonal elements report the pairwise interference between feature i and every other feature within expert e .

As Figures 1a and 2a demonstrate, at equal total parameters ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$) the dense model and the MoE represent a comparable number of features (10 versus 8) with comparable norms. The decisive difference lies not in how many features are learned but in the geometry of their interference. Although individual MoE experts exhibit some local superposition—Expert 0 in Figure 2b, for instance—the global interference structure is strictly partitioned. Whereas in the dense model (Figure 1b) any feature may interfere with any other, the MoE enforces a block-diagonal structure in which features routed to distinct experts have exactly zero interference. MoEs therefore allocate representational capacity by partitioning the feature space, sharply curtailing the global scope of interference rather than merely redistributing it.

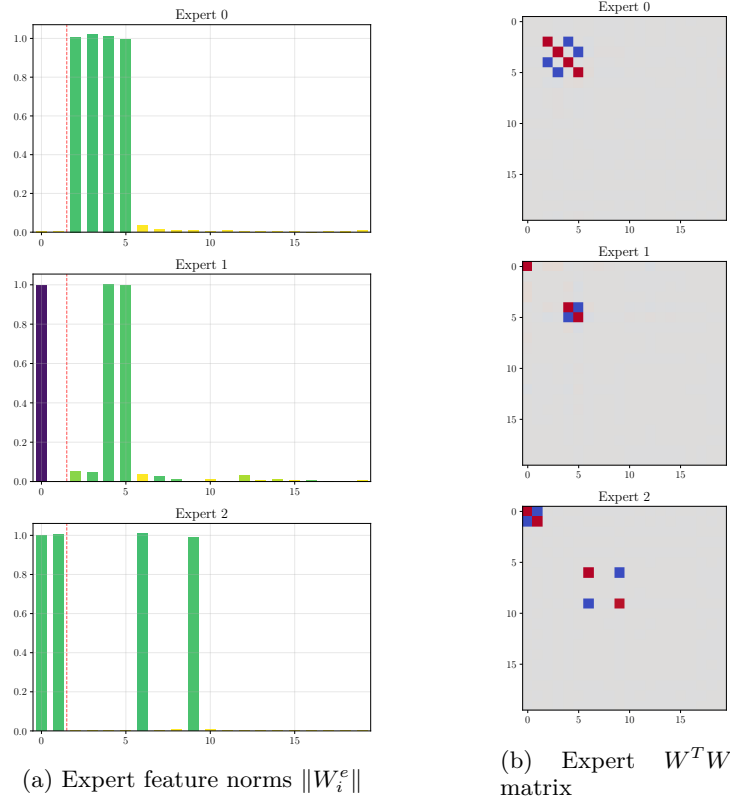


Figure 2: Feature representation and superposition in an MoE with $n = 20$ features, 3 total experts, and $m = 2$ hidden dimensions per expert under top- $k = 1$ routing, trained under exponentially decaying importance $I = 0.7^i$ and feature density $1 - S = 0.1$. Matched to the dense model of Figure 1 on total parameters, the MoE confines interference to within-expert blocks rather than distributing it across the full feature set—the block-diagonal signature of a partitioned representation.

Expert Feature Dimensionality. To characterize how an MoE allocates its limited representational capacity differently from a dense model, we measure feature dimensionality—the “fraction of a dimension” that a given feature occupies within a model (Elhage et al., 2022). For feature i , we define its dimensionality in expert e as

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

The quantity D_i^e is bounded between zero (feature not learned) and one (feature represented monosemantically), so the total representational capacity of an MoE follows naturally as $D = \sum_e \sum_{i=1}^n D_i^e$.

Efficient Packing. A model packs features “efficiently” when their dimensionalities sum to the number of embedding dimensions, $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022); for an MoE this generalizes to $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. We find empirically that both dense and MoE models satisfy this dimensionality constraint, which establishes that MoEs pack features exactly as efficiently as dense models at equal total parameters. Packing efficiency is therefore not the axis along which the two architectures differ; the distinction lies instead in how that capacity is distributed—the question we quantify next.

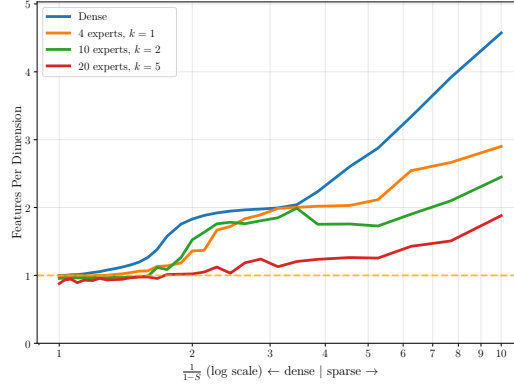


Figure 3: Features per dimension versus inverse feature density ($\frac{1}{1-S}$) for **dense** and **MoE** architectures under uniform feature importance ($I_i = 1.0$). The dense model ($n = 100$, $m = 20$) exhibits the greatest superposition, and superposition decreases monotonically as the expert count grows: 4 experts with $m = 5$, $k = 1$ (orange); 10 experts with $m = 2$, $k = 2$ (green); and 20 experts with $m = 1$, $k = 5$ (red). Because all models are matched on total parameters and on the ratio k/E , the ordering isolates the effect of network sparsity alone. The dashed line at 1.0 marks the monosemantic limit, below which the model represents no more features than it has dimensions.

Features per Dimension & Network Sparsity. Because both dense and MoE models pack features efficiently, the number of features per dimension provides a direct, architecture-agnostic measure of superposition, one that lets us quantify precisely how the expert count modulates superposition across feature sparsities. A features-per-dimension value exceeding one signals superposition, since the model then represents more features than it has dimensions. We define features per dimension for an MoE as

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where $\|W\|_F^2$ is the squared Frobenius norm and p_e is the expected probability that expert e is used across a batch of inputs—equivalently, the average renormalized gating weight after top- k routing.

At equal total parameters, the dense model registers a consistently higher features-per-dimension value than every MoE (Figure 3), and hence strictly more superposition. The dense model exploits superposition to represent more features than it has dimensions ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE caps its representation at the monosemantic limit ($n_{\text{learned}} \approx m_{\text{total}}$). This behavior is governed by network sparsity: as the total number of experts grows, with total parameters and the ratio k/E held fixed, features per dimension decreases monotonically. *The greater the number of experts, the less superposition in the model*—features become progressively more monosemantic as network sparsity increases.

Crucially, this interpretability gain is not purchased at a prohibitive cost in performance. The dense model’s additional superposition does buy it a consistently lower reconstruction loss than the MoEs, a gap of only ~ 0.03 – 0.08 at any given sparsity (Figure 6, Appendix A.1); yet as the expert count increases, the MoE loss converges toward that of the dense model, so greater monosemanticity is attained under a steadily narrowing performance penalty. Appendix A.2 develops the theoretical intuition underlying this trade-off.

5 PHASE CHANGE

Although MoEs and dense models learn a similar number of features, MoEs distribute those features across experts with substantially less interference. Network sparsity therefore reshapes how features are allocated rather than how many are learned. We now examine how the properties of the input distribution—feature sparsity and feature importance—drive this allocation, and whether they induce the physics-inspired *phase changes* that characterize dense representations.

A model commands only a finite repertoire of ways to represent any feature: it may ignore the feature, place it in superposition, or represent it monosemantically. A phase change is the observation that discrete boundaries separate these regimes as functions of feature sparsity and relative importance.

Dense toy models exhibit discontinuous ‘phase changes’ between internal feature representations (Elhage et al., 2022): varying the sparsity and relative importance of the input features elicits qualitatively distinct behavior, with greater feature sparsity, for instance, encouraging greater superposition. Analyzing the per-expert phase diagram of an MoE reveals that its experts adopt representational strategies fundamentally different from those of dense models.

Setup. We adopt the setup of Section 4.1, augmented with the auxiliary load-balancing loss to avert specialization collapse—the failure mode in which experts and routers settle into local minima that leave certain experts entirely unused. We evaluate three configurations, all with a single active expert ($k = 1$): (A) $n = 2$, $m = 1$; (B) $n = 3$, $m = 1$; and (C) $n = 3$, $m = 2$. For each, we report the per-expert phase diagram over feature sparsity and last-feature relative importance while sweeping network sparsity, increasing the expert count E up to the number of input feature dimensions n .

Here we hold active parameters fixed rather than total parameters, so that $m_{dense} = km_{moe}$ (disregarding router parameters). This choice enables a genuinely controlled comparison across architectures: were active parameters left free, any observed differences could be ascribed to that budget rather than to the architecture itself. Note that Figure 4.C.1/1 happens to match Figure 4.B.X/2 in active parameters; the latter, however, must encode the same number of input features ($n = 3$) using only a single hidden dimension ($m = 1$) rather than two ($m = 2$), which leaves little room to exploit superposition in the service of specialization.

Results and Takeaways. In every single-expert (dense) configuration we observe a clear phase change (Figure 4.X.1/1), corroborating Elhage et al. (2022). Increasing the total number of experts, however, systematically dissolves these discrete phase changes. Some experts in the $E = 2$ MoEs echo their dense counterparts (Figure 4.X.2/2) yet transition far more continuously, and in each case one expert becomes markedly more monosemantic, specializing in the feature of highest relative importance. Experts that diverge from the dense pattern uniformly exhibit much lower superposition scores—they are visibly bluer—signaling more monosemantic representations. This is precisely the behavior anticipated in the preceding section: MoEs systematically favor lower superposition than their dense counterparts.

In the $n = 2, m = 1$ configuration (Figure 4.A), the dense model declines to represent the last feature under low feature sparsity, whereas the matched MoE preserves that feature far more readily precisely because it retains the capacity to do so. With three input dimensions (Figure 4.B) this behavior vanishes: the experts already superimpose the other two features, leaving no room for the third within a single hidden dimension. Architecture B is thus the

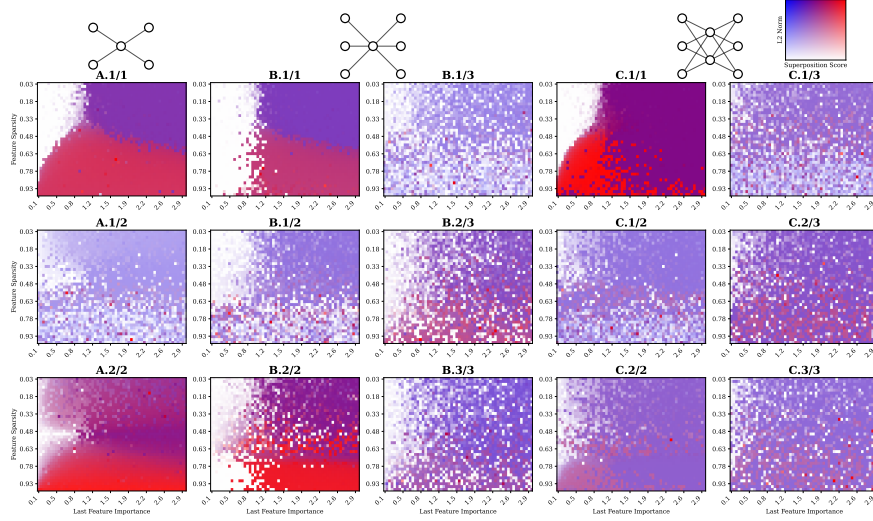


Figure 4: For a given expert and input dimension (feature), we decode how that feature is embedded in the hidden dimension—whether it is ignored (white), represented monosemantically (blue-purple), or placed in superposition (red). Each cell jointly encodes the feature norm ($\|W_n\|^2$) and the superposition score ($\sum_{j<n}(\hat{W}_n \cdot W_j)^2$) of the last feature, swept across feature sparsity $S \in [0.1, 1]$ and relative last-feature importance $I_n \in [0.1, 3]$, where the subscript n denotes the last of n total features. For every cell we train ten models and retain the one attaining the lowest loss; all runs in this section include the load-balancing loss. A low $L2$ norm ($\|W_n\|$) is rendered white to indicate that the model ignores the last feature; otherwise, a low superposition score is rendered blue-purple to indicate a monosemantic representation, whereas red indicates that the feature is held in superposition. Cell (i, j) of subfigure X.e/E reports expert e of E total experts, trained on architecture X, for last-feature importance $I_n = i$ and sparsity $S = j$; X.1/1 denotes the corresponding dense model.

exception in which the superimposing hidden dimension is, in the high-sparsity regime, insufficient to represent all features—and even here no clear phase change emerges, save for the mostly discrete but mixed band at feature sparsity 0.5–0.7 in Figure 4.B.2/2. For $m = 2$, the white region of the dense model (Figure 4.C.1/1), which occupies the mid- to low-sparsity domain where the last feature is relatively less important than the others, reflects that the feature is ignored altogether. As network sparsity increases across the remaining panels of Figure 4.C, the models instead represent the last feature with progressively greater $L2$ magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$). The dimensionality allocated in the low-importance region thus rises monotonically with network sparsity, exactly as Figure 3 predicts.

Across Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2 we observe a window of feature sparsity, roughly 0.48 to 0.7, in which polysemantic, monosemantic, and ignored representations are heavily intermixed. This intermixing marks a middle regime in which polysemantic and monosemantic representations incur comparable loss, so that the MoE cannot consistently commit to the strategies it adopts cleanly in the low- and high-sparsity domains. No analogous regime appears in dense models—direct evidence that MoEs learn qualitatively different representational strategies.

Conclusion. All of these experiments use top- $k = 1$ routing, so exactly one expert is active at any time. Even under this most dense-like of settings, we observe markedly different behavior as early as the $E = 2$ case—including when the superimposing hidden dimension has ample capacity to represent every feature. We therefore conclude that it is misleading to regard an MoE as a mere aggregation of dense models: by restricting each expert to only a subset of the feature domain, the router fundamentally alters both how the experts behave and what they learn.

6 EXPERT SPECIALIZATION

Having established that MoEs exhibit less superposition, we now turn to how these monosemantic features are organized within experts and to what that organization reveals about specialization.

Conventional treatments of expert specialization in MoEs are framed around load balancing—the even distribution of inputs across experts (Chaudhari et al., 2025). This framing, however, fails to capture the more natural intuition of specialization, under which an expert is invoked only when the concepts in which it specializes are actually present in the input.

We therefore define an expert as specialized when it *occupies* certain feature directions in the input space and represents those features relatively monosemantically. We show that these two conditions are directly correlated, and that their joint presence itself promotes load balancing across experts—thereby recovering the conventional desideratum as a consequence, rather than a definition, of specialization.

Because we fix $k = 1$, the router partitions the feature space into convex cone regions (Appendix A.3), each region routed to a single expert. Formally, $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$, where C is the set of points contained in the cone and s is any positive scalar; that is, if a feature vector $\vec{x}$ is routed to an expert, then so is every positive rescaling $s\vec{x}$. In this case we say that feature x is contained within expert e , and correspondingly that expert e *occupies* x .

We find empirically that small models which distribute the input space across more experts tend to achieve lower loss (Appendix A.4). This raises a natural question: does the allocation of the input space to particular experts entail any representational property of those experts? Our definition of specialization predicts that it entails *monosemanticity*—and, as we demonstrate below, this correlation persists in larger toy models (e.g., $m = 10$).

For models with $n > 2$, we investigate the two complementary directions of this relationship. First, we ask whether initializing an expert to occupy specific features in the input space causes it to represent those features more monosemantically. Second, and conversely, we ask whether—for the features an expert has itself chosen to represent monosemantically—that expert in fact occupies those features in the input space.

Under diagonal initialization, each expert monosemantically represents exactly the single feature it initially occupied, and no other, as shown in Figure 5a. Under ordered k-hot initialization, the first expert monosemantically represents four of the five features it initially occupied (Figure 5b); the remaining experts, initialized over less important features, represent neither those less important features nor the five most important features they were not initialized over. When we break the ordering of feature importance and randomize the features each expert initially occupies, every expert monosemantically represents only the most important feature or features among those it was initialized over, as shown in Figure 5c.

The features initially routed to an expert are thus strongly correlated with the features that expert ultimately comes to represent monosemantically. Moreover, experts represent *only* important features monosemantically—whether we initialize an expert with a single important feature explicitly or supply it a set of features from which it selects the most important of its own accord and represents that one monosemantically.

Under uniform feature importance, by contrast, experts place all features in superposition with elevated levels of interference. Even in this regime, however, the features an expert initially occupies remain *relatively* more monosemantic, attaining superposition scores that lie, on average, 1.03 standard deviations below the mean for that expert.

We next examined the converse correlation: whether an expert that represents certain features monosemantically also occupies those features in the input. To this end, we measure the expert’s usage both when its monosemantic features are *one of many* active features and when they are the *only* active features—the latter case being equivalent to measuring the probability that the expert occupies those features. As Table 1 shows, the correlation holds

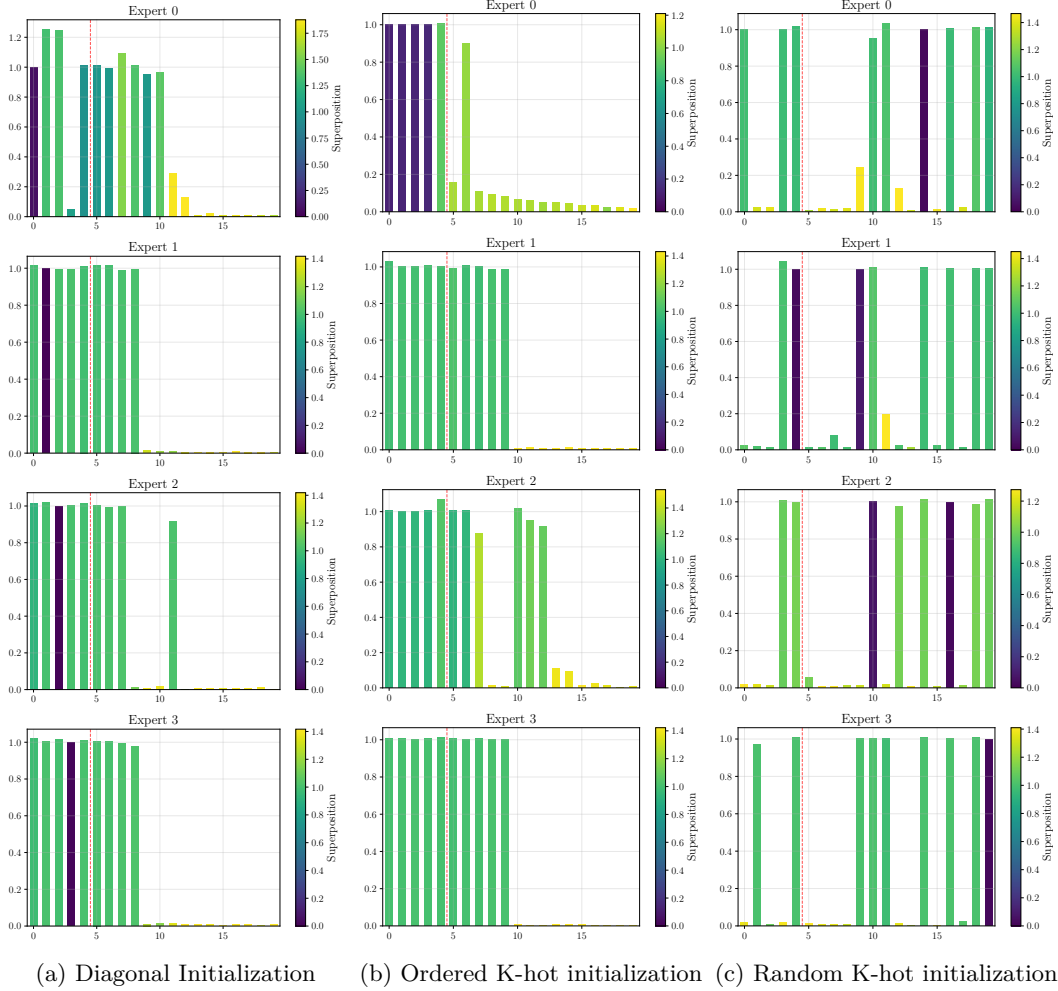


Figure 5: Expert feature norms $\|W_i^{(e)}\|$ and superposition (color) for three initialization schemes, with $n = 20, m = 5, E = 4, \hat{S} = 0.1$. In (a), the gate matrix is initialized along the main diagonal ($W_i^r = \hat{e}_i$, the basis vector for dimension i), and relative feature importance decays exponentially in order from feature one to twenty. In (b), the gate matrix is initialized to an “ordered k-hot” pattern in which the first expert aligns with the first five features and each subsequent expert with the next five; relative feature importance is as in (a). In (c), the gate matrix is initialized to a “random k-hot” pattern in which each expert is assigned five random features, with no feature shared between experts and all twenty features covered collectively; relative feature importance again decays exponentially but is now distributed randomly across features.

robustly under both Xavier and k-hot initialization. With $E = 10$, a mean expert usage of $\sim 10\%$ corresponds to even load balancing across experts. In every case, activating an expert’s monosemantic feature or features raises that expert’s usage substantially, and when those features are the sole active features the expert dominates routing outright. Under k-hot initialization the correspondence is exact: 100% of the features an expert represents monosemantically are occupied by that same expert.

An expert that represents more features monosemantically is, by our definition, more specialized: its usage on arbitrary inputs falls, yet conditional on its specialized features being active its usage rises far above that of the remaining experts. This pattern holds in every case but one—the Xavier-initialized expert representing four monosemantic features, whose

Table 1: Per-expert monosemantic-feature and usage statistics for $n = 100, m = 10, E = 10$. For each initialization scheme (Xavier and k-hot), one hundred models are trained, yielding 1000 experts per scheme. Statistics are aggregated across models, with experts grouped by the number of features they represent monosemantically. For the feature(s) an expert represents monosemantically, we report the expert’s usage when those features are one of several active features in the input and, separately, its usage when those features are the *only* active feature—the latter measuring how strongly the expert occupies those feature directions. Conditioned on the monosemantic features being active, usage rises sharply and monotonically with their number, and reaches near-total dominance when those features alone are active.

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	—	—	—
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	—	—	—
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	—	—	—
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

overall utilization drops sharply—an isolated exception set against an otherwise consistent monotonic trend spanning both initialization schemes.

7 CONCLUSION

We investigated how experts shape superposition in MoEs, establishing that MoEs consistently exhibit greater monosemanticity than dense networks while forgoing the discontinuous phase change altogether. We further proposed a feature-based definition of expert specialization and demonstrated that experts organize naturally around coherent features whenever initialization encourages such specialization. These findings rest on controlled autoencoder toy models with synthetic data, and their transfer to large-scale transformers—where the feature distribution is unknown—remains an open question (see Appendix A.6). Within these conditions, however, we show that toy MoEs attain comparable loss while sustaining markedly more interpretable representations—challenging the prevailing view that mechanistic interpretability and model capability stand in fundamental tension. Future work should determine what favors monosemanticity in MoEs, how the training dynamics of MoEs diverge from those of dense models, and when specialization emerges. Resolving these questions can directly inform the design of more interpretable, high-performing language models.

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A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

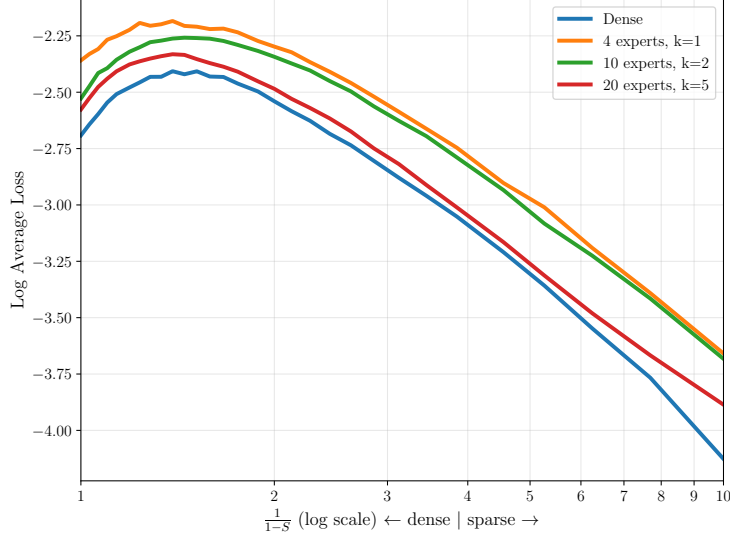


Figure 6: Log average loss versus feature density ($\frac{1}{1-S}$) for dense ($m = 20$) and MoE (4 experts, $k = 1$, $m = 5$), MoE (10 experts, $k = 2$, $m = 2$), and MoE (20 experts, $k = 5$, $m = 1$) models, all with uniform feature importance ($I_i = 1.0$) for $n = 100$ input features. Results are averaged over five runs per sparsity level. Although the dense model outperforms all MoEs at every sparsity level, the MoE loss converges toward that of the dense model as the number of experts increases.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition emerges to exploit the gap between sparse features and dense computation, compressing rarely active features into shared dimensions. MoEs, by contrast, “eat this same gap” structurally, through conditional computation: by aligning activation sparsity with feature sparsity, they eliminate the computational penalty otherwise incurred by dedicated, rarely activating neurons. As Elhage et al. (2022) observe, once a model expends computation only on active features, splitting polysemantic neurons into dedicated monosemantic ones becomes the optimal strategy—in effect trading the *compression* of superposition for the *selection* afforded by routing.

This structural change manifests geometrically as a reduction in *interference*. Although the global ratio of represented features to parameters remains constant, the router effectively partitions the feature space, ensuring that a feature routed to expert e competes for capacity only with the subset of features also assigned to that expert. Interference is consequently governed by the local expert matrix $(W^e)^T W^e$ rather than by the global $W^T W$ of a dense model. This partitioning sharply reduces the number of interfering features for any given feature vector, relieving the optimization pressure to pack features in superposition and thereby permitting experts to learn monosemantic representations.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

In the regime $k = 1$, the router function $g(x) = \text{softmax}(W^r x)$ is equivalent to $\text{argmax}(W^r x)$. The region routed to expert i can then be written as $\forall j \neq i, (w_i - w_j)^T x > 0$, where w_i and w_j are row vectors of W^r . This is a homogeneous linear inequality, and regions bounded by such inequalities are by definition convex cones: if a particular x satisfies

the inequality, then so does sx for any positive scalar s ; and if x_1 and x_2 both satisfy it, then so does every convex combination $x = \lambda x_1 + (1 - \lambda)x_2$ for $\lambda \in [0, 1]$.

When $k > 1$, the region of inputs routed to a particular expert e becomes a union of convex cones. Because the union of convex cones is not in general itself a convex cone, the characterization of experts as occupying feature directions need not extend beyond $k = 1$.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

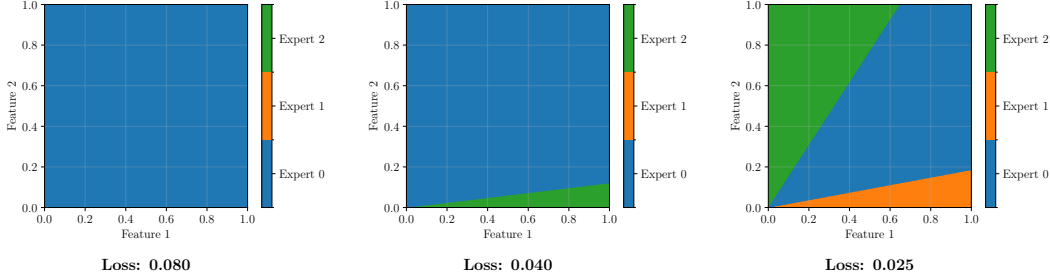


Figure 7: Expert routing of three identical models under differing initialization schemes, with $n = 2$, $E = 3$, $m = 1$. The first model (left) attains the worst performance (loss: 0.08) and routes all inputs to a single expert. The second model (middle) performs better (loss: 0.04), routing a small portion of inputs—specifically those for which feature 1 is active—to a second expert. The third model (right) achieves the lowest loss (loss: 0.025) and distributes the input space across all experts: one expert is chosen when only feature 1 is active, one when only feature 2 is active, and one when both are active.

In small models ($n = 2$, $m = 1$, $E > 1$), and sweeping $E \in [2, 7]$, we find empirically that models distributing the input space across more experts tend to achieve lower loss. Fixing $n = 2$ allows us to visualize directly which portions of the input space are routed to which experts, as shown in Figure 7.

A.5 ANALYTIC MODEL EQUIVARIANCE

For the toy setup of single-layer, single-nonlinearity, top- $k = 1$ MoEs, there exists a theoretical map between any dense model and a monosemantic MoE representing an equivalent number of active features under a sparsity constraint.

Assume there exists an upper bound a on the number of active features for any input, so that $\forall x \in D: |\{i : x_i \neq 0\}| \leq a$. Assume further that a is no greater than the hidden dimensionality m of an expert, which bounds the number of features the model must represent, and that this hidden dimensionality is in turn smaller than the total number of input features n (i.e. $a \leq m \leq n$). To construct the monosemantic MoE, for each subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$ —that is, each subset of active features of size at most a —we create an expert that monosemantically preserves those features. (It in fact suffices to take only the subsets with $|S| = a$.) The router then selects the expert corresponding to the active features, of which, by assumption, there are never more than a :

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where $\text{support}(x) = \{i : x_i \neq 0\}$. Since $|S| \leq a \leq m$, each expert possesses sufficient capacity to represent its assigned features without superposition. To restate the construction: at most a features are active, and every distinct combination of active features receives its own dedicated expert with capacity sufficient to represent those features monosemantically. The number of experts required is therefore $\binom{n}{m}$.

The reconstruction of this theoretical MoE attains zero loss only insofar as the sparsity constraint holds (equivalently, as sparsity approaches one in these toy models), since more

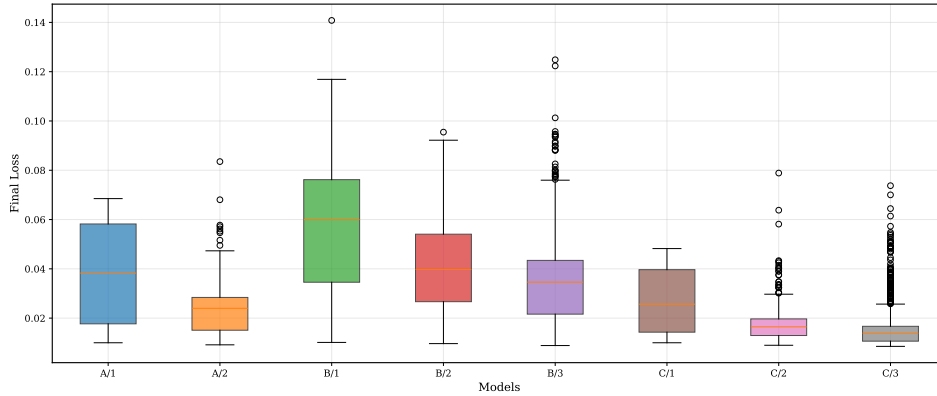


Figure 8: Model X/E uses X to denote the same model architectures used in Figure 4 and E to denote the total number of experts (i.e. network sparsity). Increasing network sparsity decreases mean loss while increasing localized variance—especially as the expert count approaches the number of input feature dimensions. We attribute this variance to the relatively unstable training of MoEs compared with dense models, which persists despite our training ten models per cell and retaining the lowest-loss run.

than m features could in principle be active simultaneously ($a \not\leq m$), which would exceed the monosemantic representational capacity of the network. Even then, however, the dense polysemantic model could do no better unless features are correlated in the distribution: whenever $a \not\leq m$, the polysemantic model confronts the identical obstacle, so the monosemantic MoE under this construction may still outperform it under looser sparsity constraints.

Thus, for any dense model $f_{\text{dense}}(x) = \text{ReLU}(Wx+b)$ operating under the sparsity constraint $|\text{support}(x)| \leq a$, there exists an MoE model $f_{\text{MoE}}(x)$ such that $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for all valid inputs. In the toy settings of this paper, the sparsity constraint holds in the limit where sparsity approaches one. In practice, moreover, the number of features that a given quantity of information can semantically encode may itself be bounded—as indicated by the dimensionality of meaningful embeddings of that data—so that an MoE with sufficiently many experts and a tractable, interpretable degree of superposition may suffice to encode every feature present.

A.6 LIMITATIONS

Our results should be interpreted in light of several limitations that also delineate the scope of our claims. First, all experiments are conducted in controlled toy-model settings derived from superposition studies in sparse autoencoders (Elhage et al., 2022). While this design affords precise measurement of interference and monosemanticity, it abstracts away many complexities of large-scale transformers—including multiple layers of computation, attention, heterogeneous feature distributions, and realistic routing dynamics—so that the transferability of our findings to production MoE architectures remains to be established.

Second, our architectural and routing choices are intentionally restricted: we use simple experts, fixed top- k routing (most often $k = 1$), equal parameter budgets between dense and MoE models, and no auxiliary load-balancing losses. Practical MoEs frequently employ richer routing mechanisms, variable expert capacities, and multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022), or extend beyond reconstruction to objectives such as next-token prediction, any of which may yield qualitatively different representational behaviors.

Third, the feature distributions of our synthetic tasks—their sparsity patterns, feature-importance profiles, and independence assumptions—are considerably simpler than those encountered in natural data.

Finally, although we observe increased monosemanticity and reduced superposition in MoEs under fixed conditions, we do not evaluate downstream performance trade-offs, training stability, or specialization dynamics at scale; prior work indicates that such dynamics can shift substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

In sum, our study delivers mechanistic insight under clean, controlled experimental conditions, and validating these patterns in large, realistic MoE systems is the natural next step. Because our present methods do not scale directly to such models, developing scalable measurements of superposition constitutes a clear and compelling direction for future work.